PINZ AND MUHAMMAD
MASS ALL ROUNDER

CS101 – Introduction to Computing
GDB Problem Statement

The government is planning to launch a National Social Media App aimed at promoting responsible information sharing among users and controlling the spread of wrong information. The app will specifically include the following features:

Users must register using their CNIC Number for authentication.
The platform allows users to share verified information through posts on the app.
Multiple users have the option to react to a post, expressing their feedback through likes or comments.
Users can initiate real-time chats by sending chat requests to each other, facilitating interactive communication on the platform.
You are required to suggest a network model that can be used for the mentioned Social Media App. You are asked to choose from the following network models. Also, provide a brief and to-the-point justification in support of your preferred choice.

1- **Client/Server model**

2- **Peer-to-Peer model**

In the Client/Server model, a singular process acts as the client, making requests to another process designated as the server. On the other hand, in the Peer-to-Peer model, processes mutually offer and avail services from one another.

Solution:-

For the National Social Media App aimed at promoting responsible information sharing and controlling the spread of wrong information, the recommended network model is the **Client/Server model.**

Justification:

1.Centralized Control:The Client/Server model offers a centralized architecture where a dedicated server manages and controls the flow of information. This aligns with the government's goal of promoting responsible information sharing as it allows for better monitoring and control over the content shared on the platform.

2.Authentication and Security: With the requirement for users to register using their CNIC (Computerized National Identity Card) number for authentication, the Client/Server model is well-suited. The server can handle user authentication centrally, ensuring a secure and verified user base.

3.Verified Information Sharing: The platform's focus on sharing verified information is facilitated by the central server. The server can implement processes to verify and authenticate the information before it is allowed to be posted on the platform, enhancing the reliability of the shared content.

4.Real-Time Communication: The ability for users to initiate real-time chats aligns with the interactive features of the Client/Server model. The server can facilitate and manage real-time communication between users efficiently.

5. Efficient Resource Management: The Client/Server model allows for efficient resource management as the server can handle tasks like authentication, verification, and content control. This ensures a smoother and more controlled user experience.

In summary, the Client/Server model provides the necessary centralized control, security, and efficient management features required for a social media platform with a focus on responsible information sharing.